HW 5

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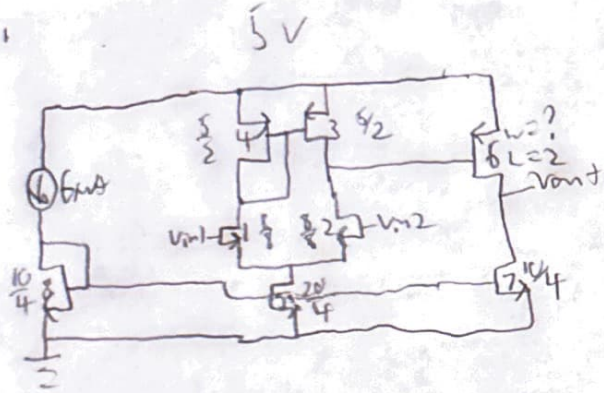
EIEN 252

1.



$K_n' = 180$ $\lambda_n = .05$ $V_T = .6$

$K_p' = 120$ $\lambda_p = .02$

a. Vin1 is inverting
Vin2 is noninverting

b. 12mA through 5, 6 through 1,2,3,4
6 through 7 and 6
b/c the currents are equal and are a pair

$W_6 = 5$

c. $V_{I(M} > V_{ov5} + V_{GS1}$

$V_{I(M} < V_{DD} - V_{SG4} + V_T$

$12mA = \frac{1}{2}(180\times10^{-6})\frac{20}{4}(V_{ov5})^2$

$V_{ov5} = .1633V$

$6mA = \frac{1}{2}(180\times10^{-6})\frac{5}{8}V_{ov1}^2$

$V_{ov1} = .2582V$

$V_{GS1} = V_{ov1} + V_T = .8582$

$V_{I(M\,min} = .8582 + .1633 =$ $1.0215V$

$6mA = \frac{1}{2}(120\times10^{-6})\frac{5}{2}V_{ov4}^2$

$V_{ov4} = .2V$

$V_{SG4} = .2 + .6 = .8V$

$V_{I(M\,max} = 5 - .8 + .6 =$ $4.8V$ maybe $4.2V$

e. $R_{out} = r_{o6} \| r_{o7} =$ 2380952.38Ω

d. $A_1 = g_{m1}(r_{o2} \| r_{o3})$ $A_T = A_1 \cdot A_2$

$A_2 = -g_{m6}(r_{o6} \| r_{o7})$

$g_{m1} = \frac{2(6\times10^{-6})}{.2582} = 4.6476\times10^{-5}S$

$V_{ov6} = V_{ov4} = .2$

$g_{m6} = \frac{2(6\times10^{-6})}{.2} = 6\times10^{-5}S$

$r_{o2} = r_{o3} = \frac{1}{(6\times10^{-6})(.05)} = 3333333.\bar{3}\Omega$

$r_{o3} = r_{o6} = \frac{1}{(6\times10^{-6})(.02)} = 8333333.\bar{3}\Omega$

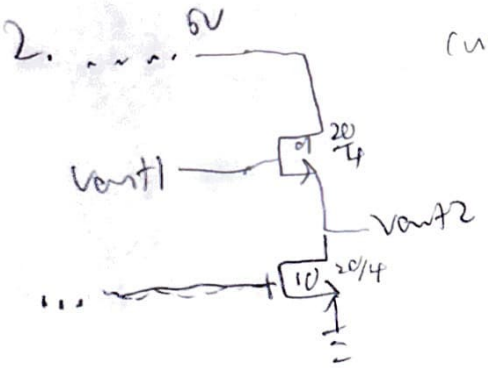
$r_{o2} \| r_{o3} = r_{o6} \| r_{o7} = 2380952.38\Omega$

$A_1 = 4.6476\times10^{-5}(2.38M\Omega) = 110.6567\,V/V$

$A_2 = 6\times10^{-5}(2.38M\Omega) = -142.857\,V/V$

$A_T = -15808.095\,V/V$

2.



$curr = 6\times10^{-6}(2) = 12\times10^{-6}$

$$A_{v_{sf}} = g_{m9}\left(\overbrace{r_{o10} \| r_{o9} \| \frac{1}{g_{m9}} \| \frac{1}{g_{mb9}}}^{R_{out}}\right)$$

$$12\times10^{-6} = \frac{1}{2}(180\times10^{-6})\frac{20}{4}V_{ov}^2$$

$$V_{ov9} = V_{ov10} = .1633V$$

$$g_{m9} = g_{m10} = \frac{2(12\times10^{-6})}{.1633} = 1.4697\times10^{-4}S$$

$$g_{mb9} = 1.4697\times10^{-5}S$$

b. $r_{o9} = r_{o10} = \frac{1}{12\times10^{-6}(.05)} = 1666666.\bar{6}\Omega$

$$r_{o10} \| r_{o9} \| \frac{1}{g_{m9}} \| \frac{1}{g_{mb9}} = \boxed{6140.0048\Omega = R_{out}}$$

c. $1.4697\times10^{-4}(6140.0048) = .9024 V/V$

$$A_{v_T} \cdot A_{sf} = \boxed{-14265.1101 V/V}$$